



L LOVELY
P PROFESSIONAL
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A
PROJECT REPORT

On

**(AUTO PARKING SPACE VISION DETECTOR USING
IMAGE PROCESSING)**

Submitted to

LOVELY PROFESSIONAL UNIVERSITY (LPU), PUNJAB

In partial fulfillment of the requirements for the award
of the Degree

**BACHELOR OF COMPUTER APPLICATIONS
(LATERAL ENTRY)**

Submitted By:

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CERTIFICATE

This is to certify that the project entitled “**Auto Parking Space Vision Detector using image Processing**” submitted by **Uvesh Ahmad (22106330700)**, in partial fulfillment of the requirement for the award of the **Bachelor of Computer Applications (Lateral Entry)**. They have worked under my guidance and Supervision and have fulfilled all the requirements for the submission.

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DECLARATION

This to certify that the project report titled “**Auto Parking Space Vision Detector using image Processing**”, submitted by **Uvesh Ahmad, Registration No. 22106330700**. In partial fulfillment for the award of **Bachelor of Computer Applications (Lateral Entry)** from **Lovely Professional University, Phagwara, Punjab**, is a record of original work done by me. This project has not been submitted to any other University or Institution for the award of any Degree.

Signature of the Student

Place:

Date:

ACKNOWLEDGEMENT

We are highly indebted to the project Supervisor **Ms. Pallvi Arora** and all other respected Professor of Computer Engineering, Lovely Professional University Phagwara, Punjab for enabling us to work on such a project. Their constant encouragement and valuable guidance has been very precious. We are thankful to them for all the formal and informal discussion, constructive criticism and valuable suggestions which were a guiding factor in continuous improvement in project. We are highly thankful to the friends and colleagues who gave the required confidence and support that always helped in boosting the morale and provide great and much needed motivation.

Place: New Delhi,

Date: 05 December, 2023

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ABOUT THE PROJECT

Creating an “**Auto Parking Space Vision Detector using image Processing**” in **Python** involves leveraging **Computer Vision** techniques to analyze a given area for available parking spaces. Initially, I capture the video feed or use a static image of the parking lot.

The image preprocessing step is crucial, where I convert the image to grayscale, apply filters like Gaussian blur to enhance edges, and reduce noise. Next, object detection techniques come into play; I did use contour detection to identify individual parking spaces, segment them from the background using thresholding, and filter contours based on specific criteria like size or shape. Once spaces are detected, I did keep track of their occupancy status (occupied or vacant) and implement a counting mechanism based on this status. Visualizing the detected parking spaces, marking them as 'occupied' or 'vacant', and displaying the real-time count of available spaces is essential.

This process can involve creating a loop to continuously process video frames, implement an alert system when spaces reach a certain threshold, and possibly save data on parking occupancy over time for analysis. The code in **Python** utilizing **OpenCV** libraries would handle tasks like accessing video feed, performing counting logic within the loop, and displaying results or updating counters while allowing for user interruption. Fine-tuning parameters and rigorous testing across various parking scenarios will be crucial for accuracy in detecting and counting parking spaces.

Overall the project is to detect the **Auto Parking Space Vision Detector sing image Processing**, which allow us to act accordingly, by that we can count the space parking lot.

CHAPTER 1

INTRODUCTION

This chapter provides an introductory glimpse into the innovative solution, Auto Parking Space Vision Detector using image Processing (APSVDuIP). It introduces the system while delving into the problem statement, objectives, scope, and the project report structure for a comprehensive understanding.

This section serves multiple purposes. It begins by contextualizing the field or topic under discussion, highlighting its relevance, importance, or the challenges it faces. For instance, in the context of image processing, it might emphasize the increasing reliance on digital imagery across various industries and the need for efficient processing techniques. Furthermore, the introduction outlines the general layout of the document, giving a concise overview of the key topics or sections that will be explored without delving into specific details. It provides a roadmap, signaling what readers can expect in subsequent chapters.

1. 1 Introduction on Image Processing

Image Processing involves defining an image as a two-dimensional function, $i(x, y)$, where x and y represent spatial coordinates on a plane. The amplitude of 'f' at any coordinate pair (x, y) signifies the intensity or gray level of the image at that specific point. If x , y , and the intensity values of 'f' are all finite and discrete, it transforms the image into what is known as a digital image. The realm of digital image processing revolves around manipulating these digital images using a digital computer.

In essence, a digital image comprises a finite set of elements, each possessing a distinct location and value. These elements go by various names such as picture elements, image elements, pels, or most commonly, pixels. The term 'pixel' widely represents the fundamental elements constituting a digital image.

There a Formula summarizing the essence of the explanation provided:

- **$i(x, y)$** : Represents the Two-Dimensional function defining the image where **x** and **y** denote spatial coordinates.

Image: $i(x, y)$ for spatial coordinates (x, y)

- $i(x, y)$: Represents intensity at each (x, y) .
- Finite, discrete x , y , and intensity create a digital image.
- Pixels: unique elements in the digital image.
- Digital image processing manipulates pixels via computer.

1.2 Introduction on the System

Nowadays, car is very important to everyone especially for who are works. People are willing to make installment to get own car. When talking about metropolitan, then traffic jam always occur because of numbers of vehicles are so high. Thus we cannot deny the existing of the cars in -our daily life. Whenever we go out by car, we are facing problem to find an available parking space due to the tremendous increase of occupancy of cars.

The analogy is when driver enterla certain parking lot, the first thing that the driver do is looking forward of some sign to telling that the parking lot is fully occupied, partly occupied -or vacant. The driver also do not know how many are there and where to find a parking -division for his/her car. Some of parking divisions may -' remain unoccupied even the total occupancy is high. This will causing ineffective use of parking divisions as well as traffics jams around the entrance of parking lot. Therefore, by offering drivers with relevant information on the parking lot during entering a parking lot becomes an important issue.

The proposed system called as Auto Parking Space Vision Detector using Image Processing. This system proposes a method of detecting the existence of parked vehicles by processing the image of the parking lot taken by a surveillance camera and then counting the available parking space which is display in front of entrance of parking lot. The system employ images, since all area in the parking lot can be observed with relatively few camera. Other than that, the system is compact and the cost is not is not expensive. The image of a parking lot is taken by a surveillance camera set at some height in the parking lot.

1.3 Problem Statement

There are some reasons why Auto Parking Space Vision Detector using Image Processing is developed. The problems that have been identified are stated below:

- i. Driver needs some relevant information before entering the parking lot such as the current available parking spaces in the parking lot.
- ii. There are current system used in parking lot but the method used is based on the detection by installing a certain sensor on each division; the other is to detect cars through images of the parking lot taken by surveillance cameras. In the method with the sensor, the cost rises as the number of parking divisions because a lot of sensors are required corresponding to each parking divisions.

- iii. Driver might be takes time to find available parking space in parking lot. Imagine if the parking lot has many number of parking divisions and driver will through all parking divisions just to find an available parking space.

1.4 Objective

Objectives of the Auto Parking Space Vision Detector using Image Processing to be developed are to:

- i. Capture and detect existence of vehicle at parking lot using image processing technique.
- ii. Count, display available parking space and the location of the available parking spaces in parking lot.

1.5 Scope of the Project

There are a few scopes that have been identified in order to develop the system. The scopes of the systems are:

- i. This system is just a prototype system using image processing techniques.
- ii. Using image that captured from external webcam and using model simulation.
- iii. The position of the parked vehicle is correct.
- iv. The location of The Parking Space Counter techniques can be readily implemented at Lovely Professional University's parking lot.
- v. The system can be used in daytime only without have a strong shadow.

CHAPTER 2

LITERATURE REVIEW

This chapter will briefly defines some similar ideas about current system, existing system, technique or algorithm that used and comparison on advantages and disadvantages between existing systems. The literature review section of a document critically evaluates and synthesizes existing scholarly works, studies, and internet relevant to the subject matter. This section aims to provide an in-depth analysis, comparison, and discussion of the available literature, highlighting key findings, methodologies, theories, and gaps in the field. It serves to contextualize the current state of knowledge, identify trends, contradictions, and areas requiring further exploration or research within the domain being studied.

2.1 Project definition

Auto Parking Space Vision Detector using Image Processing is prototype system that detecting the existence of parked vehicles by processing the image of a parking lot taken by surveillance camera. After that, system will count how many available parking lot spaces each division and display it in front of entrance parking lot.

Image Processing defines as analysis of a picture using techniques that can identify shades, colors and relationships that cannot be perceived by the human eye. Image processing is used to solve identification problems, such as in forensic medicine or in creating weather maps form satellite pictures. It deals with images in bitmapped graphics format that have been scanned in or captured with digital cameras. Image Processing also defines as any image improvement, such as refining a picture in a paint program that has been scanned or entered from a video source.

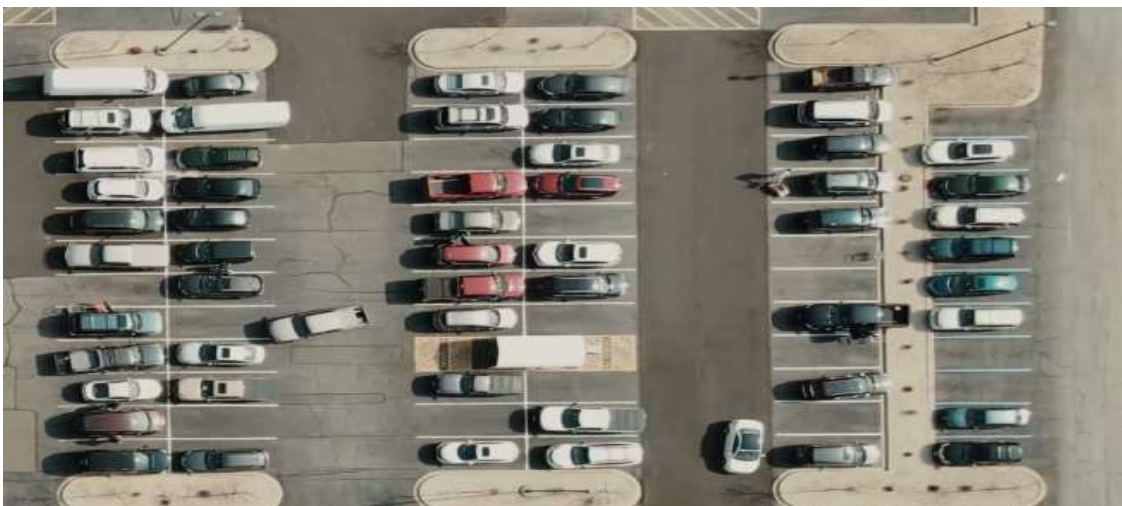


Figure 2.1 Example of Real Image Parking Lot

Figure 2.1 shows the example of parking lot. In this figure, there are available parking spaces to drivers parking their cars. This prototype system will process the image to get available parking space in parking lot.

2.2 Studies of Existing Systems

For research purpose, here are stated four examples of the applications to be compared in term of operational features. Below are list of similar existing systems:

- i.** Car-Park Occupancy Information System (COINS).
- ii.** Locating Vehicle in a Parking Lot by image Processing.
- iii.** Parking Guidance System using LCD and Image Processing Technique in WSN Environment. Parking Space Vacancy Monitoring.

2.2.1 Comparison between Car-Park Occupancy Information System (COINS) and Locating Vehicle in a Parking Lot by Image Processing

Nowadays, drivers always face difficulty of finding available parking lot while entering into a huge parking area. A Car-Park Occupancy Information System (COINS) is developed to be a viable solution to reduce the amount of time needed to search for a vacant car-park lot especially in a huge parking area. With this system, images captured by a surveillance camera were processed in real-time to identify the occupancies of the parking lots. This occupancy information is further processed by a central control unit and distributed to display panels located at strategic locations at the parking area. The drivers can easily find a vacant parking lot based on the information displayed on the panels. Motivation for developing this system came from the fact that minimum cost is involved because image processing technique is used rather than sensor-based techniques. As surveillance cameras are readily available in most car parks, this technique is much cost effective than installing sensor on each parking lot.

Locating Vehicle in a Parking Lot by image Processing is more concern to propose a method of detecting the existence of parked vehicles by processing the image of a parking lot taken by a surveillance camera. Whenever driver wants to park a car at a parking lot, how to find a proper parking division there causes a serious problem. The objective of the present article is in providing drivers with such information as the lot is fully occupied or relatively vacant, where unoccupied parking divisions are found, and so on. The images employed, since all areas in the parking lot can be observed with relatively few cameras, the system is compact, and the cost is not expensive. The image of a parking lot is taken by a surveillance camera set at some height in the parking lot.

The relevant issues are how to cope with both temporal and spatial changes in illumination, how to discriminate shadows from vehicles, how to cope with occlusion, and how to cope with various surface reflectance of vehicles and so on. To cope with these issues, the input images transformed to the gray levels with log transform, extracts edges and counts the number in each parking division, and then decides if each division is occupied or not. The recognition rates for a set of images taken at various moments of a day were well above 95 %.

2.2.2 Comparison between Parking Guidance System using RFID and Image Processing Techniques in WSN Environment and Parking Space Vacancy Monitoring

Parking Guidance System using RFID and Image Processing Techniques in WSN Environment describes a novel approach to developing a Parking Guidance System within the car park in an environment in order to help alleviate the frustration and problem in finding vacant parking space. The system utilizes the existing CCTVs installed in the car park coupled with FPGA device in detecting the vacant spaces which will in turn be assigned to the patron using the shortest path algorithm based on both the point of entrance to the car park and building. The patron is then guided to the specified location by referring to the map printed on the parking ticket. Besides that, an RFID tag is also attached to the parking ticket to uniquely identify the assigned parking space of the patrons and will be used to remind patrons of the parking location during payment.

Whereas, Parking Space Vacancy Monitoring is propose a stereo-vision based system that deal with instances with severe vehicular occlusion. In this system, multiple cameras are used to monitor the vacancy status of the P502 parking spaces on more places.



Figure 2.2: An example of severe vehicular occlusion in the scale model

Figure 2.2 shows the example of severe vehicular occlusion in the scale model. In this system, a method for monitoring vacancies in parking lots using a stereo camera system presented to create a 3D reconstruction of the scene, which enables us to determine the vacancy status of a particular parking space under vehicular occlusion. Additionally, results for 3D reconstruction using uncalibrated versus calibrated cameras are compared. This system is able to identify vacancies while differentiating between spaces for different permit holders (faculty versus students). Ideally, the system also able to provide an exact count of the number of available spaces, but it must have to appeal to a statistical notion of vacancy, as certain spots may be too heavily occluded by trees, other vehicles, etc. to be monitored with very high accuracy. This information will ultimately be integrated with a status dissemination tool, where drivers will be able to query the parking lot status via mobile phone.

2.3 Studies Approach, Method and Technique on Existing System

Studies Approach

The Studies Approach, also known as the research design, is the overarching framework that guides the entire research process. It encompasses the fundamental decisions that determine the nature, scope, and direction of the study. The Studies Approach outlines the following key elements:

- i. **Research Question or Hypothesis:** Formulating a clear research question or hypothesis provides the central focus of the study and sets the direction for data collection and analysis.
- ii. **Research Design:** Selecting an appropriate research design, such as experimental, quasi-experimental, observational, or correlational, determines the structure and methodology of the study.
- iii. **Sampling:** Determining the sampling strategy, including sample size, sampling method, and participant demographics, ensures the representativeness and generalizability of the findings.
- iv. **Data Collection Methods:** Identifying the appropriate methods for collecting data, such as surveys, interviews, observations, or experiments, ensures that the data gathered is relevant and reliable.
- v. **Data Analysis Techniques:** Selecting appropriate data analysis techniques, such as statistical analysis, qualitative coding, or content analysis, allows for the systematic interpretation of the collected data.

Method

The Method, also known as the research procedure, outlines the specific steps involved in carrying out the research plan. It provides a detailed roadmap for conducting the study and ensures consistency and replicability of the research process. The Method typically includes the following components:

- i. **Operational Definitions:** Defining key variables and concepts clearly and unambiguously ensures that all participants and researchers have a shared understanding of the terms used in the study.
- ii. **Data Collection Procedures:** Providing detailed instructions and procedures for data collection ensures that the data is gathered in a standardized and reliable manner.
- iii. **Instrumentation:** Describing the tools and instruments used for data collection, such as questionnaires, observation checklists, or experimental apparatus, provides transparency and allows for assessment of the validity and reliability of the data.
- iv. **Data Management Plan:** Outlining a plan for data storage, organization, and protection ensures the integrity and security of the collected data.

Technique

The Technique refers to the specific tools, skills, or procedures employed to implement the research method. It represents the practical application of the research plan and directly contributes to the collection and analysis of data. The Technique can be further categorized into:

- i. **Data Collection Techniques:** Specific methods used to gather data, such as survey questions, interview techniques, or observation protocols, represent the techniques employed in data collection.
- ii. **Measurement Techniques:** Tools and procedures used to measure variables or assess constructs, such as standardized tests, rating scales, or coding rubrics, represent the techniques employed in data measurement.
- iii. **Data Analysis Techniques:** Specific statistical procedures, qualitative coding methods, or content analysis techniques used to interpret and draw conclusions from the collected data represent the techniques employed in data analysis.

2.3.1 Capture image

Image capture is the process of acquiring an image of a scene or object using a camera or other imaging device. This is the first step in many image processing tasks, as it provides the raw data that will be used for subsequent analysis.

There are many different ways to capture an image, depending on the specific needs of the application. Some common methods include:

- i. **Using a digital camera:** This is the most common method of image capture. Digital cameras use a sensor to convert light into an electronic signal, which is then stored as an image file.
- ii. **Using a webcam:** Webcams are small cameras that are typically used for video conferencing. They can also be used to capture still images.
- iii. **Using a scanner:** Scanners are used to capture images of physical objects, such as documents, photographs, and artwork.
- iv. **Using a frame grabber:** Frame grabbers are used to capture images from video sources, such as video cameras and surveillance systems.
- v. Once an image has been captured, it can be stored, processed, and analyzed using a variety of image processing techniques.

2.3.2 Feature extraction and point matching

This step involves identifying and extracting key features from the image. These features can be anything from the edges of objects to the color of pixels. Point matching is a technique that is used to find corresponding points in two or more images. This can be used to track objects or to create a 3D model of a scene.

2.3.4 Object Detection for vehicle

This step involves identifying and locating objects in an image. Object detection is a challenging task because objects can vary in appearance and can be occluded by other objects.

Object detection is a computer vision technique that involves identifying and locating objects in an image or video. Object detection is a challenging task because objects can vary in appearance, can be occluded by other objects, and can be present in different lighting conditions.

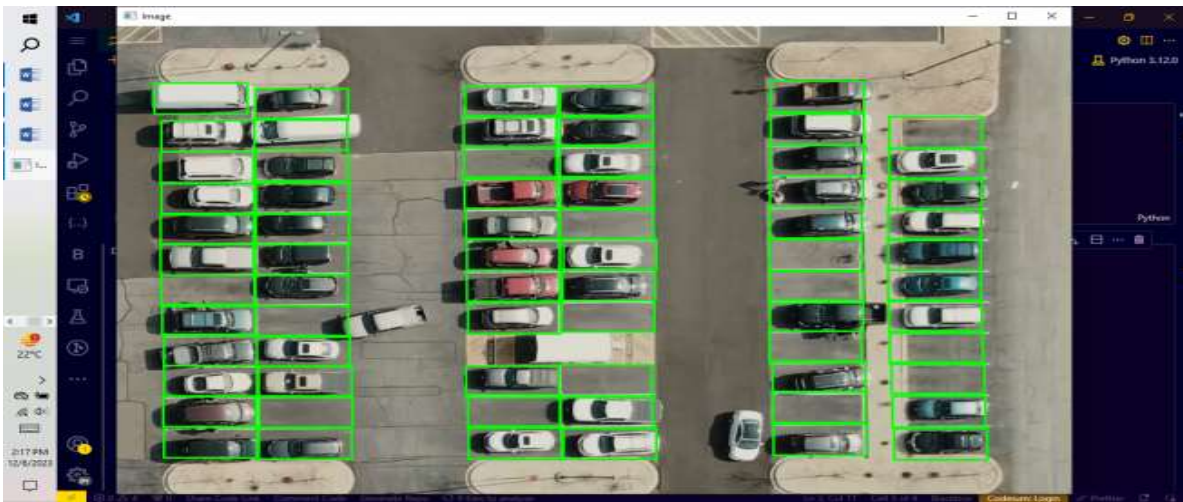


Figure 2.3.4: An apply Object Detection Scale

There are many different object detection algorithms, but some of the most common ones include:

- i. **Viola-Jones object detection:** Viola-Jones object detection is a relatively simple and fast object detection algorithm that is based on like features.
- ii. **YOLO (You Only Look Once):** YOLO is a high-performance object detection algorithm that uses a convolutional neural network (CNN) to identify objects in an image.
- iii. **SSD (Single Shot Multi Box Detector):** SSD is another high-performance object detection algorithm that uses a CNN to identify objects in an image.

2.4 Component of an Image Processing Systems

The key components of an image processing system for parking space counting include image capture, preprocessing, feature extraction, object detection, data processing, decision making, user interface, and system integration. These elements collectively enable the system to capture, process, and analyze data, facilitating accurate counting and assessment of parking space availability within an area.

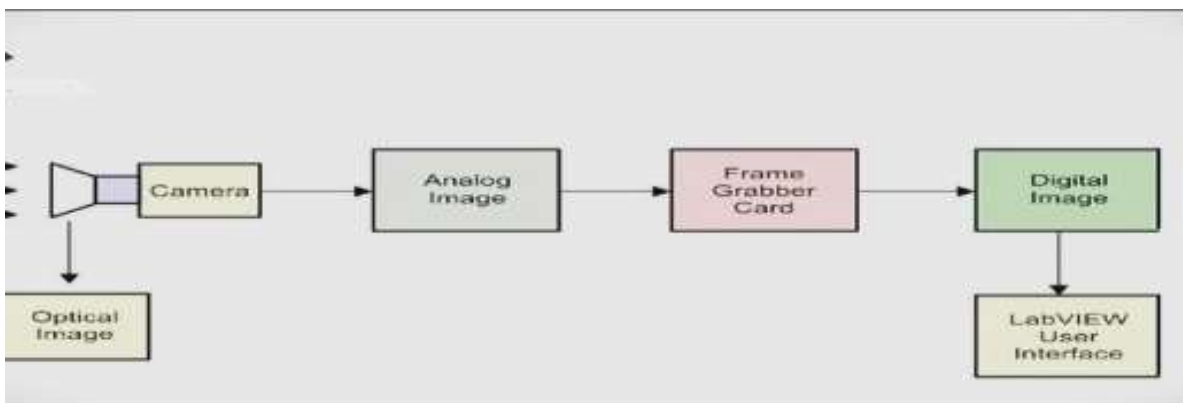


Figure 2.4: An example of Image Processing Systems Model

An image processing system is a computer system that is designed to process and analyze images. The main components of an image processing system are:

- i. **Image acquisition:** This is the process of capturing an image.
- ii. **Image preprocessing:** This is the process of preparing an image for analysis.
- iii. **Feature extraction:** This is the process of identifying and extracting key features from an image.
- iv. **Image classification:** This is the process of assigning an image to a particular class.
- v. **Image segmentation:** This is the process of dividing an image into meaningful segments.
- vi. **Image reconstruction:** This is the process of reconstructing an image from incomplete or noisy data.

CHAPTER 3

METHODOLOGY

Introduction

The Software Development Life Cycle (SDLC) has been widely adapted through most, if not all, technology based companies. With the advancement of technology, and the increasing number of companies that rely on their own custom applications, learning and understanding the complexities of the SDLC is increasingly important. Of course knowing how to code is the basis for any software engineering career, but understanding the SDLC brings a lot of new skill bases to the table, all of which are beneficial to the engineering process. These skills include planning, designing, and testing. The SDLC traditionally has ten phases which are initiation, system concept development, planning, requirements analysis, design, development, integration and testing, implementation, operations and maintenance, and disposition. Despite being distinct phases, they are not all required in every situation and often are either skipped or combined together. These phases are all addressed in more detail later. Additionally, variations on the SDLC, including increasingly important aspects devoted to security in software engineering, are addressed as well.

3.1. The Software Development Life Cycle (SDLC)

The Software Development Life Cycle is a basis or framework for structuring, planning, and executing tasks involved with developing an information system. Due to the difference in requirements of systems along with numerous other differences, variations on the SDLC have been developed. Before beginning a project, it is important to determine which of these is the most suitable, especially with the recent advancement of web-based technologies. The SDLC includes distinct phases designed to give software engineers a clearly defined goal for how to work. It aims to aid developers and other project staff to create a system that meets all technical and user requirements as well as exceeds customer expectations. These phases all revolve around a central theme of quality assurance. The major priority of the project team is to provide quality software to the users and client. A visual of this cycle is demonstrated below.

SDLC ensures a systematic, well-structured process that helps manage risks, control costs, and ensure the end product meets user needs and quality standards. It's a tried-and-true method for creating reliable and effective software systems.



Figure 3.1: SDLC Systems Model

The SDLC typically consists of the following phases:

1. **Planning:** The planning phase involves defining the scope of the project, identifying the stakeholders, and creating a project plan. This phase lays the foundation for the entire software development process and ensures that everyone involved is on the same page.
2. **Analysis:** The analysis phase involves gathering and understanding the requirements of the software. This includes identifying the users, defining the use cases, and creating a requirements document. This phase ensures that the software meets the needs of the users and addresses all of their functional and non-functional requirements.
3. **Design:** The design phase involves creating a blueprint for the software. This includes designing the architecture of the software, creating user interfaces, and developing data models. This phase translates the requirements from the analysis phase into a technical plan that can be implemented by the developers.
4. **Development:** The development phase involves coding the software. This includes writing unit tests, integrating the code, and testing the software. This phase brings the software to life by transforming the design into a functional application.

5. **Testing:** The testing phase involves verifying that the software meets the requirements and is free of defects. This includes unit testing, integration testing, and system testing. This phase ensures that the software is bug-free, meets the user's expectations, and adheres to the established requirements.
6. **Deployment:** The deployment phase involves releasing the software to the users. This includes installing the software, training the users, and providing ongoing support. This phase marks the transition of the software from development to production and ensures that the users can successfully utilize the application.

3.2 Integration and Testing

The Integration and Testing phase aims to determine if the requirements specified in the specifications document are met. Three different types of testing are ideal to include in this phase: integration testing of subsystems, security testing, user testing or acceptance testing, and unit testing. Each of these types of tests serve a different purpose and help all stakeholders determine flaws in the system prior to deployment. Unit testing, which tests small portions of code and is traditionally done during the development phase. Several analysis reports may be produced from these tests and should be presented to stakeholders before moving on to the implementation phase. During this meeting, details on how implementation should take place are imperative to address.

3.3 Implementation

During this phase, the system is deployed to the production environment. It is important to inform users of the potential system changes prior to deployment and to train any professionals on the new system if required. A post-implementation review is commonly completed to make sure all requirements are met and the system is functioning in production as expected. If everything is in order, a contract between the development team and the client project manager is signed, stating that the system has been completed and delivered.

3.4 Operations and Maintenance

More than half the cost of the development life cycle is due to the Operations and Maintenance phase. This phase may include identifying system operations, maintaining data, identifying problems, and revising documentation that received additional analysis (e.g. security audit/analysis). A User-Satisfaction report may be developed during this phase to identify and potential user end problems to address in future development iterations.

3.5 Object Classification

Object classification has seen significant advancements in recent years, driven by the progress in deep learning and artificial intelligence. Here are some key developments:

1. **Improved Accuracy:** Deep learning models, particularly convolutional neural networks (CNNs), have achieved remarkable accuracy in object classification tasks. Pre-trained models like VGG16, ResNet, and Inception have become commonplace for high-performance object recognition.
2. **Increased Efficiency:** Lightweight model architectures and efficient training techniques have led to faster inference times and lower hardware requirements. This enables real-time object classification on mobile devices and edge computing platforms.
3. **Multi-Object Detection:** Modern algorithms can detect and classify multiple objects within an image simultaneously, enhancing the capabilities for scene understanding and analysis.
4. **Object Localization:** Bounding boxes or segmentation masks can now accurately localize objects in images, providing valuable information for tasks like robot vision and autonomous navigation.
5. **Domain-Specific Adaptation:** Deep learning models can be fine-tuned for specific domains, such as medical imaging or industrial inspection, achieving superior performance on specialized datasets.
6. **Open-Source Platforms:** Numerous open-source tools and libraries, like TensorFlow, PyTorch, and OpenCV, enable developers to easily build and deploy object classification systems.
7. **Explainable AI** Efforts are underway to develop explainable AI techniques, providing insights into the decision-making process of deep learning models. This increases trust and transparency in the application of AI for object classification.

3.6 System Requirement

The specific system requirements for object classification depend on the intended application and desired functionalities. However, some general requirements include:

Hardware:

- i. **Processing Power:** CPUs and GPUs with sufficient processing power for running deep learning models efficiently.
- ii. **Memory:** Adequate RAM to store large pre-trained models and datasets.

iii. **Storage:** Enough storage space for training data, model weights, and processed results.

Software

- i. **Operating System:** A stable operating system compatible with the chosen deep learning framework.
- ii. **Deep Learning Framework:** TensorFlow, PyTorch, or other frameworks for building and training object classification models.
- iii. **Libraries and Tools:** Additional libraries like OpenCV for image manipulation and libraries specific to the chosen framework.

Data

- i. **Dataset:** A high-quality dataset containing labeled images for training and evaluating the object classification model.
- ii. **Data Preprocessing:** Tools and techniques for pre-processing images before feeding them to the model.

Performance Metrics

- i. **Accuracy:** The percentage of correctly classified objects.
- ii. **Precision:** The proportion of true positives among all positive detections.
- iii. **Recall:** The proportion of true positives among all actual positives.
- iv. **Inference Speed:** The time it takes to classify an object in real-time.

Security

- i. Secure storage and handling of sensitive data.
- ii. Protection against adversarial attacks and model poisoning.

CHAPTER 4

TESHNOLOGIES USED

4.1 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) illustrates the flow of data within a system or a process. In the context of "Counting Available Parking Space Using Image Processing," here's how you might structure the DFD:

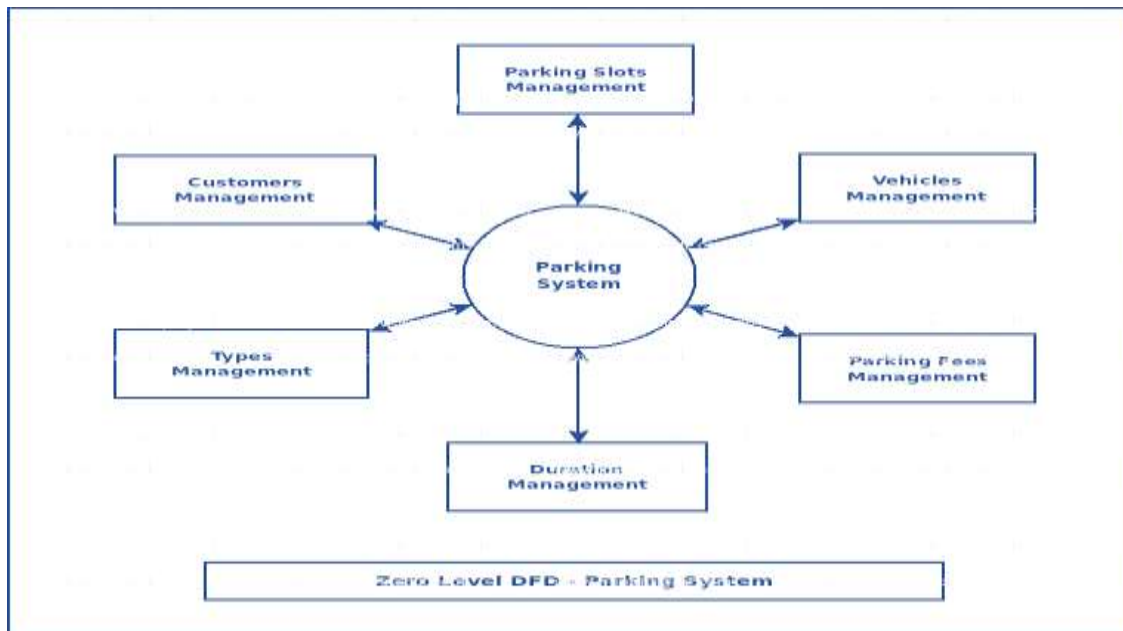


Figure 4.1: DFD Diagram Model

Data Flow Diagram (DFD) for Counting Available Parking Space Using Image Processing
Context Diagram

Parking Lot System: Represents the entire system of counting available parking spaces using image processing.

Input: Image data captured from surveillance cameras.

Output: Processed information about available parking spaces.

Level 0 DFD

Processes:

Image Capture

Image Processing for Parking Space Counting

Output Generation

Data Flows:

Input Image Data

Processed Parking Space Information

System Parameters (if any)

Level 1 DFD (Detailed Processes)

Image Capture Process:

Input: Raw image data from surveillance cameras.

Output: Raw image data for processing.

Image Processing for Parking Space Counting:

Sub-processes:

Feature Extraction, Thresholding, Object Detection, Counting Algorithm

Input: Raw image data.

Output: Processed parking space information.

Input: Processed parking space information.

Output: Display of available parking spaces or data transmission to user interface.

Data Stores

Raw Image Data Store: Storage for captured images.

Processed Data Store: Storage for processed parking space information.

External Entities, Surveillance Cameras: External devices providing image data.

User Interface: External interface displaying available parking space information to users.

The DFD visualizes the flow of data through various processes in the system. It delineates how image data is captured, processed, and ultimately used to generate information about available parking spaces. Each process and data flow represents a distinct part of the system's functionality in counting available parking spaces using image processing techniques.

4.2 ER Diagram

An Entity-Relationship (ER) diagram is a visual representation of the relationships between entities within a system or a database. It's a conceptual modeling technique used in software engineering and database design to illustrate the logical structure of data and the connections between different entities.

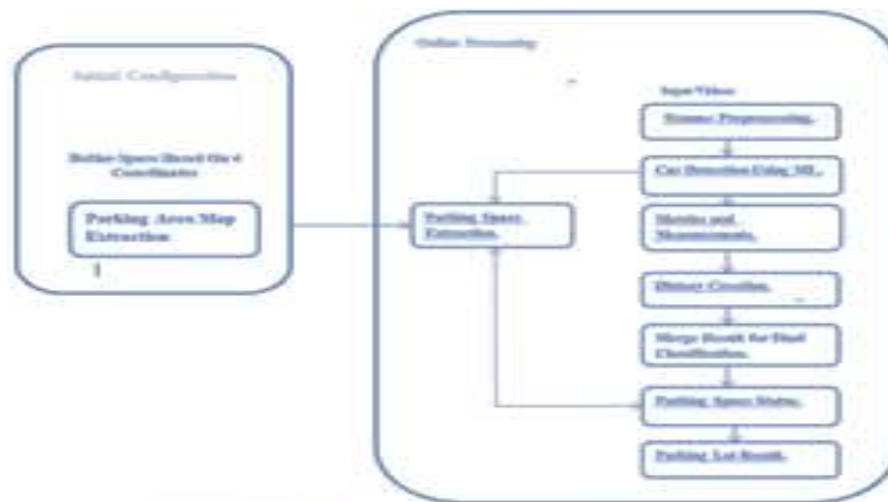


Figure 4.2: Proposed system architecture

Figure 4.2: ER Diagram Model

4.2.1 Components of an ER Diagram:

Entities: Represent the fundamental objects or things in the system, such as a person, place, thing, or concept. Entities are usually nouns and have attributes that describe them.

Attributes: Characteristics or properties that describe entities. For example, for an entity "Person," attributes might include name, age, and address.

Relationships: Depict how entities are connected or linked to one another within the system. Relationships demonstrate how data is shared between entities.

4.3 Image Processing

Image processing is a technique used to analyze and manipulate digital images. It involves the use of mathematical algorithms and computer software to perform operations on images, such as filtering, enhancement, and restoration. Image processing techniques are used in a variety of fields, including medicine, robotics, and surveillance. Some common applications of image processing include image recognition, object detection, and image restoration. Image processing is an important field that plays a significant role in many modern technologies and applications.

4.4 Python Notebook

Jupyter Notebook is an open-source web application that allows users to design and share documents that contain live code, equations, visualizations, and narrative text. It supports a

Variety of programming languages, including Python, R, and Julia, and provides an interactive Computing environment for data analysis, scientific research, and education. Jupyter Notebook allows users to write and execute code in small, reusable blocks called cells, which can be easily Modified and tested. It also provides a range of visualization tools for exploring data and communicating insights. Jupyter Notebook has become a popular tool in the data science community due to its flexibility, ease of use, and support for reproducible research.

4.5 Python

Python is a famous, general-purpose programming language known for its simplicity, readability, and flexibility. It is a high-level language, which means that it is easier to learn and use than lower-level languages like C or Assembly. Python is used in various types of applications, such as data analysis, scientific computing, web development and artificial intelligence

4.6 Keras

Keras is a powerful neural networks API written in Python. It is designed to be user-friendly, modular, and extensible, making it easy to build and experiment with deep learning models. Keras provides a simple and intuitive interface for building various types of neural networks, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and generative adversarial networks (GANs). It also supports multiple backends, including TensorFlow, Theano,

and CNTK, allowing developers to use Keras with their preferred backend. Overall, Keras is a powerful and popular tool for building deep learning models.

4.8 OpenCV

OpenCV (Open Source Computer Vision Library) is an open source machine learning library used for computer vision. OpenCV was built to provide a common source for computer vision applications and to accelerate the use of machine recognition in the commercial products. OpenCV is an Apache licensed product, so it makes easy for businesses to utilize and modify the code. The library contains more than 2500 optimized algorithms, which includes a

Comprehensive set of both classic and state-of-the-art computer vision and machine learning algorithms.

4.9 Pandas

Pandas is a dynamic data manipulation and analysis library for Python. It provides a wide range of functions and tools for working with structured data, including data cleaning, filtering, grouping, merging, and reshaping. Pandas data structures, such as the Series and DataFrame, are highly efficient and can handle large datasets with ease. Additionally, Pandas supports a wide range of file formats, including CSV, Excel, SQL, and more, making it easy to import and export data from various sources. Pandas is widely used in data science, finance, economics, and many other fields where data analysis is a critical task.

5.0 Numpy:

NumPy is a Python library for scientific computing and numerical analysis. It provides a set of powerful tools for working with arrays, matrices, and other multi-dimensional data structures. NumPy is built on top of low-level mathematical libraries, making it fast and efficient for numerical operations. Some of the key features of NumPy include array manipulation, indexing, slicing, sorting, and linear algebra operations. NumPy is widely used in various domains, including machine learning, data science, and scientific computing. Overall,

NumPy is a fundamental library for numerical computing in Python and is an essential tool for many data scientists and researchers.

CHAPTER 5

IMPLEMENTATION

During this phase, the system is deployed to the production environment. It is important to inform users of the potential system changes prior to deployment and to train any professionals on the new system if required. A post-implementation review is commonly completed to make sure all requirements are met and the system is functioning in production as expected. If everything is in order, a contract between the development team and the client project manager is signed, stating that the system has been completed and delivered.

5.1 Image Feature Extraction

Image feature extraction is the process of identifying and extracting key features from an image. These features can be anything from the edges of objects to the color of pixels. Feature extraction is an important step in many image processing tasks, such as object detection, image classification, and image segmentation.

There are many different algorithms for image feature extraction, but some of the most common ones include:

- i. **Scale-invariant feature transform (SIFT):** SIFT is a feature extraction algorithm that is invariant to scale, rotation, and illumination changes.
- ii. **Speeded-up robust features (SURF):** SURF is a feature extraction algorithm that is similar to SIFT, but it is faster and more robust to noise.
- iii. **Histogram of oriented gradients (HOG):** HOG is a feature extraction algorithm that is often used for object detection. It is based on the idea that the distribution of oriented gradients in an image can be used to distinguish between different types of objects.
- iv. The specific feature extraction algorithm that is used will depend on the specific task that is being performed. For example, if the task is to detect vehicles in an image, then SIFT or SURF would be a good choice. However, if the task is to classify images of flowers, then HOG would be a better choice.
- v. In this project, we will use a combination of color subtraction, thresholding, and edge detection to extract features from the images of parking lots.

5.2 Conversion of Image into Double Type

The first step in the implementation is to convert the image from an integer type to a double type. This is necessary because the image processing algorithms that we will be using require that the images be in double precision.

5.3 Normalization of ROB image

The next step is to normalize the image. This means that we will adjust the brightness and contrast of the image so that all of the pixels are within a certain range. This is important because it will make the image easier to process.

5.4 Thresholding the image

We will then threshold the image. This means that we will convert all pixels that are above a certain threshold to white and all pixels that are below the threshold to black. This will help to further isolate the objects in the foreground.

Thresholding is a fundamental image processing technique used to create a binary image from a grayscale or color image. It involves converting an image into a binary format, where pixels are categorized as either foreground (objects of interest) or background based on a certain threshold value.

- i. **Grayscale Conversion:** If the image is not already in grayscale, it's converted to a single-channel grayscale image. Each pixel in this grayscale image represents the brightness or intensity of the corresponding pixel in the original image.
- ii. **Threshold Selection:** A threshold value is chosen. This value serves as a boundary: pixels with intensities above this threshold are classified as foreground, while those below are considered background.
- iii. **Pixel Classification:** Each pixel's intensity in the grayscale image is compared to the threshold. If the intensity is greater than the threshold, it's set to a maximum value (often 255 in an 8-bit image), representing the foreground. If the intensity is lower, it's set to a minimum value (often 0), representing the background.
- iv. **Binary Image Creation:** As a result of this comparison, a new binary image is formed where pixels classified as foreground are represented as white (or 1) and those as background are represented as black (or 0).

5.6 Threshold The Gray image

Thresholding is a common technique used in image processing to separate objects from the background. It involves converting each pixel in an image to either black or white



Figure 5.6 Threshold Gray Image

Based on its intensity value. In this case, we apply thresholding to the grayscale image obtained from the previous step. The threshold value is chosen carefully to distinguish between the foreground (vehicles) and the background (parking lot).

5.7 Dilating and Filling the image

Dilation is an operation that expands the boundaries of objects in an image. It is often used to connect nearby objects and make them more visible. In this project, dilation is applied to the thresholded image to ensure that all connected pixels belonging to a vehicle are included.

Filling is another operation that complements dilation. It fills any holes or gaps within the objects, making them more complete. This helps in accurately identifying and counting vehicles.

5.8 Regrayscale and Closing the Image

After dilation and filling, the image is converted back to grayscale to maintain consistency with the previous steps. Closing is applied to the grayscale image to further solidify the objects and remove any remaining noise or small gaps. This refinement improves the accuracy of vehicle detection.

5.9 Edge Detection

Edge detection is a technique that identifies the boundaries of objects in an image. It highlights the transitions between different intensity levels, making it easier to distinguish objects from their surroundings. In this project, edge detection is applied to the closed grayscale image to enhance the contours of vehicles and make them more prominent for further processing.

5.10 Find Minimum Value

After edge detection, the image may contain isolated edge pixels or noise that could interfere with vehicle detection. To minimize these artifacts, the minimum value of the edge detection result is identified and used to threshold the image once again. This step helps to remove any unwanted noise and refine the edge detection results.

5.11 Object Classification

The refined edge-detected image is then subjected to object classification. This involves applying machine learning algorithms to identify and classify the objects present in the image. The specific algorithm used depends on the complexity of the objects and the desired level of accuracy. In this case, a trained object detection model is used to recognize vehicles based on their characteristic features and edges.

5.12 Interface Counting Available Parking Space using Image Processing Technique

The final step involves implementing a user-friendly interface that displays the processed image and the count of detected vehicles. This interface allows for real-time monitoring of available parking spaces and provides a convenient way to manage parking resources. The image processing techniques described above contribute to the accuracy and reliability of the parking space detection system.

By combining these image processing techniques, the system can effectively identify and count vehicles in parking lots, providing valuable information for parking management and occupancy monitoring.

CHAPTER 6

CODE OF THE REQUIREMENT

The code of the requirement (COR) is a concept in systems engineering that defines the specific functional and non-functional requirements for a system or component. It acts as a bridge between the system's high-level goals and the detailed technical specifications that guide implementation.

6.1 Components of COR:

- i. **Functional Requirements:** These define the specific functions and features that the system must perform. They typically describe what the system should do, not how it should achieve it.
- ii. **Non-Functional Requirements:** These specify the performance, quality, and other characteristics of the system. They typically address how the system should perform, not what it should do.
- iii. **Constraints:** These are limitations or restrictions that the system must adhere to. They can be technical, physical, or environmental in nature.
- iv. **Verification Criteria:** These define the tests or methods used to ensure that the system complies with the requirements.

6.2 Benefits of COR:

- i. **Improved Communication:** COR provides a clear and concise document that facilitates communication between stakeholders, developers, and testers.
- ii. **Reduced Errors:** By explicitly defining the requirements, COR helps to avoid misunderstandings and ambiguity that can lead to errors in design and implementation.
- iii. **Increased Efficiency:** COR can help to streamline the development process by providing a structured framework for defining and managing requirements.
- iv. **Enhanced Traceability:** COR enables traceability between requirements, design documents, and test cases, making it easier to track the progress of the project and identify potential issues.

6.3 Different Types of CORs:

- i. **User Requirements:** These capture the needs and expectations of the users of the system.
- ii. **System Requirements:** These define the overall functionality and behavior of the system.
- iii. **Subsystem Requirements:** These specify the requirements for individual components or subsystems within the larger system.
- iv. **Interface Requirements:** These define the interactions between different parts of the system or with external systems.

6.4 Importing libraries

Some Import Code in python languages and screen shorts.

Car Parking Space Counter

```
[ ]:
*([4]): # Import the OpenCV library for image and video processing
import cv2
# Import the 'pickle' module for serializing and deserializing Python objects
import pickle

# Import the OpenCV library for image and video processing
# Import the 'pickle' module for serializing and deserializing Python objects
# Import cvzone, a wrapper library built on top of OpenCV for additional functionalities
# Import the NumPy library for numerical computing
import cv2
import pickle
import cvzone
import numpy as np
```

Figure 6.4 Screen Short Code

6.5 Loading the images

```
# Set initial values for width and height
width, height = 400, 40

try:
    # Attempt to open a file named 'carparking' for reading in binary mode
    with open('carparking', 'rb') as f:
        # Load data from the file using pickle and store it in posList
        posList = pickle.load(f)
except:
    # If the file cannot be opened or data cannot be loaded, set posList as an empty list
    posList = []

# Define a function to handle mouse click events
def onMouseClick(event, x, y, flags, param):
    if event == cv2.EVENT_LBUTTONDOWN: # Check if left mouse button is clicked
        # Append the coordinates of the mouse click to posList
        posList.append([x, y])

    if event == cv2.EVENT_RBUTTONDOWN: # Check if right mouse button is clicked
        # Iterate through posList and remove a point to it until within a specified area
        for i, pos in enumerate(posList):
            x1, y1 = pos
            if x1 < x + 10 & width and y1 < y + 10 & height:
                posList.remove(pos)

    # Save the updated posList to a file named 'carparking' using pickle
    with open('carparking', 'wb') as f:
        pickle.dump(posList, f)

# Define a loop to continuously display the image and handle mouse events
while True:
    # Read an image named 'carparking.png'
    img = cv2.imread('carparking.png')

    # Draw rectangles on the image based on the positions stored in posList
    for pos in posList:
        cv2.rectangle(img, pos, (pos[0] + width, pos[1] + height), (0, 255, 0), 2)

    # Display the image to a window named 'image'
```

```
# Continuous loop for processing video frames
while True:
    # Reset video to the beginning if it reaches the end
    if cap.get(cv2.CAP_PROP_POS_FRAME) == cap.get(cv2.CAP_PROP_FRAME_COUNT):
        cap.set(cv2.CAP_PROP_POS_FRAME, 0)

    # Read a frame from the video feed
    success, img = cap.read()

    # Convert the frame to grayscale
    imgGray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Apply Gaussian blur to the grayscale image
    imgBlur = cv2.GaussianBlur(imgGray, (3, 3), 0)

    # Apply adaptive thresholding to create a binary image
    imgThreshold = cv2.adaptiveThreshold(imgBlur, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
                                         cv2.THRESH_BINARY_INV, 25, 28)

    # Apply median blur to the thresholded image
    imgMedian = cv2.medianBlur(imgThreshold, 5)

    # Define a kernel for image dilation
    kernel = np.ones((3, 3), np.uint8)

    # Apply dilation to the median-blurred image
    imgDilate = cv2.dilate(imgMedian, kernel, iterations=1)

    # Check parking space occupancy using the processed image
    checkParkingSpace(imgDilate)

    # Display the original image with parking space indicators
    cv2.imshow('image', img)

    # Wait for a specified time (20 milliseconds) for a key event
    cv2.waitKey(20)
```

Figure 6.5 Loading the images

6.6 Code Text Base

```
# Import the OpenCV library for image and video processing
# Import the 'pickle' module for serializing and deserializing Python objects
import cv2
import pickle
```

```

# Video feed
# Open the video file 'carPark.mp4' for video processing
cap = cv2.VideoCapture('carPark.mp4')

# Load previously stored positions from 'CarParkPos' file using pickle
with open('CarParkPos', 'rb') as f:
    posList = pickle.load(f)

# Define the width and height for parking space detection
width, height = 107, 48

# Function to check parking space occupancy
def checkParkingSpace(imgPro):
    spaceCounter = 0

    # Iterate through the stored positions
    for pos in posList:
        x, y = pos

        # Crop the image using the defined width and height
        imgCrop = imgPro[y:y + height, x:x + width]

        # Calculate the number of non-zero pixels in the cropped image
        Count = cv2.countNonZero(imgCrop)

        # Determine parking space status based on the count of non-zero pixels
        if count < 950:
            color = (0, 0, 255) # Red color for occupied space
            thickness = 3
            spaceCounter += 1
        else:
            color = (0, 255, 0) # Green color for free space
            thickness = 2

    # Draw rectangles on the original image to represent parking spaces
    cv2.rectangle(img, pos, (pos[0] + width, pos[1] + height), color, thickness)
    # Add text displaying the count of non-zero pixels on parking spaces

```



```

cvzone.putTextRect(img, str(count), (x, y + height - 3), scale=1,

                    thickness=2, offset=0, colorR=color)

# Add text displaying the number of free spaces out of total defined spaces
cvzone.putTextRect (img, f'Free: {spaceCounter}/{len(posList)}', (100, 50), scale=2,
                    thickness=4, offset=15, colorR = (0, 200, 0))

# Continuous loop for processing video frames
while True:
    # Reset video to the beginning if it reaches the end
    if cap.get(cv2.CAP_PROP_POS_FRAMES) == cap.get(cv2.CAP_PROP_FRAME_COUNT):
        cap.set(cv2.CAP_PROP_POS_FRAMES, 0)

    # Read a frame from the video feed
    success, img = cap.read()

    # Convert the frame to grayscale
    imgGray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Apply Gaussian blur to the grayscale image
    imgBlur = cv2.GaussianBlur(imgGray, (3, 3), 1)

    # Apply adaptive thresholding to create a binary image
    imgThreshold = cv2.adaptiveThreshold(imgBlur, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
                                         cv2.THRESH_BINARY_INV, 25, 16)

    # Apply median blur to the thresholded image
    imgMedian = cv2.medianBlur(imgThreshold, 5)

    # Define a kernel for image dilation
    kernel = np.ones((3, 3), np.uint8)

    # Apply dilation to the median-blurred image
    imgDilate = cv2.dilate(imgMedian, kernel, iterations=1)

    # Check parking space occupancy using the processed image
    checkParkingSpace(imgDilate)

```

6.7 Output Screen Shorts



Figure 6.7.1 Real Image before implementation



Figure 6.7.2 Image Processing

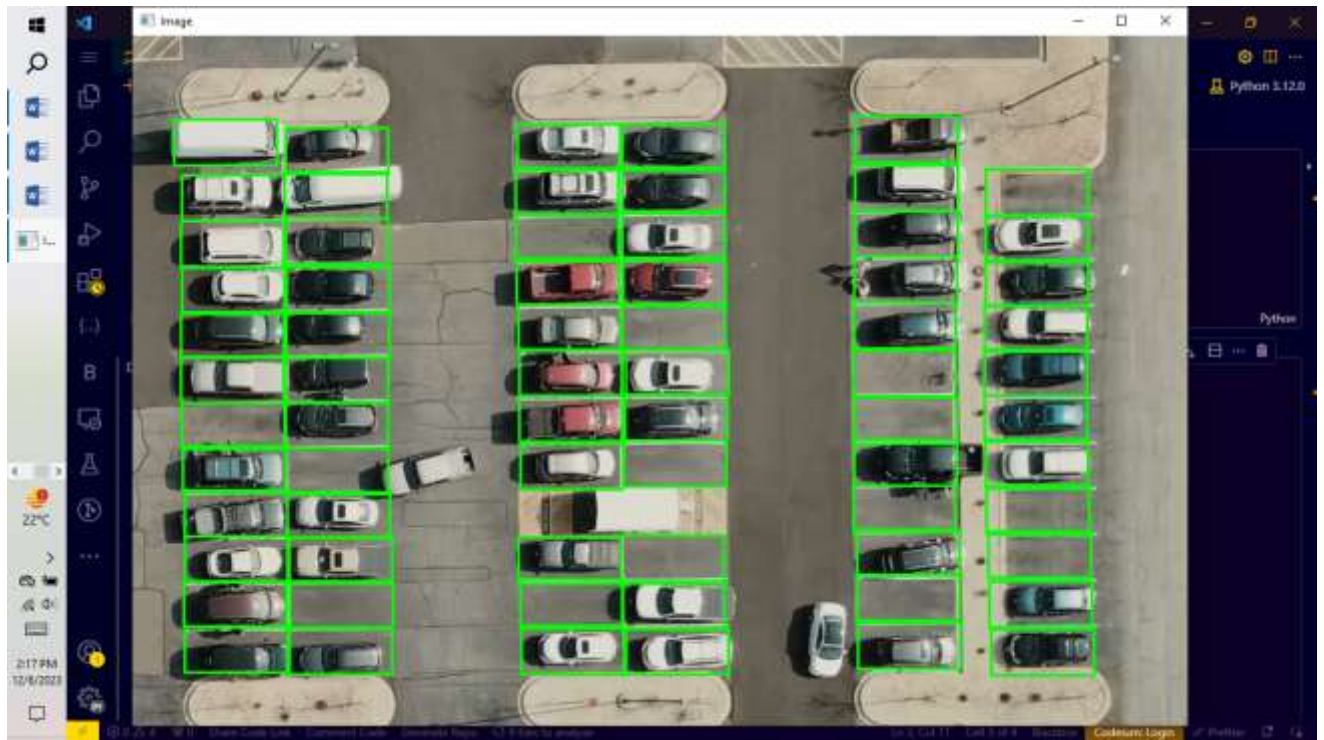
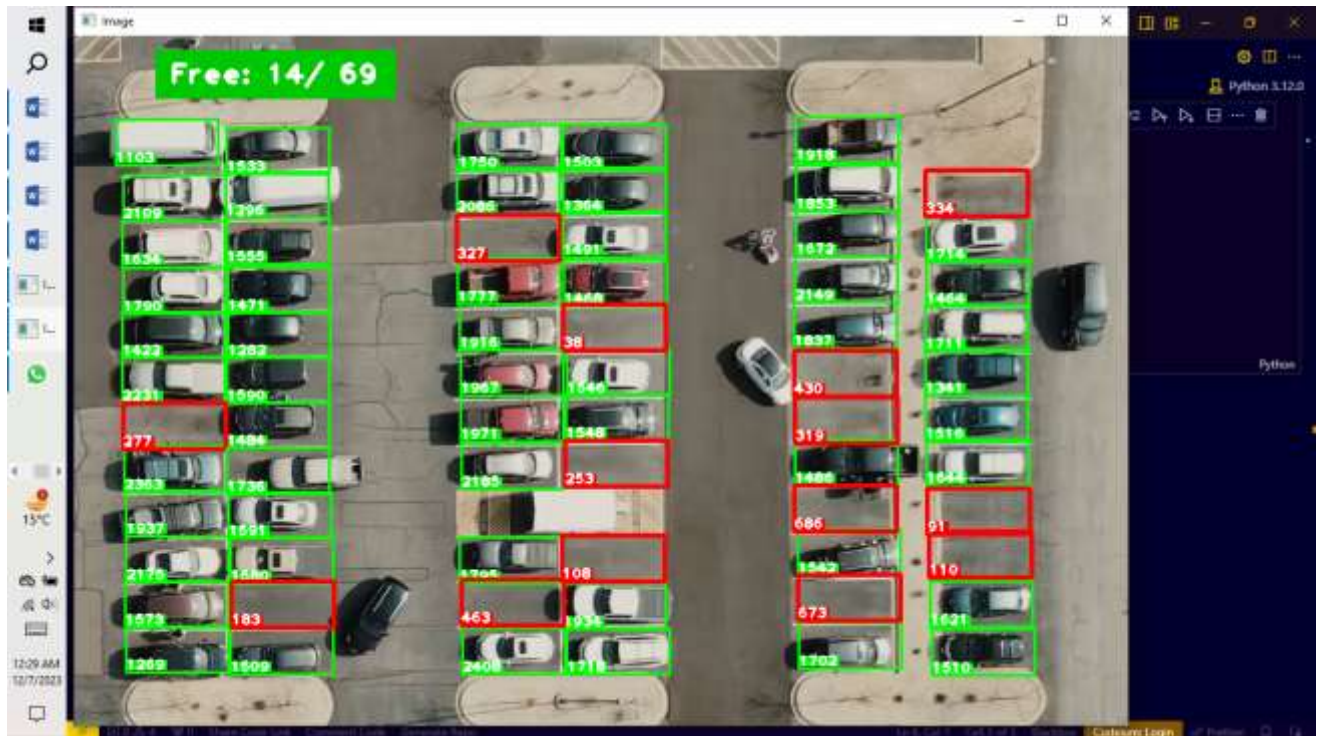


Figure 6.7.3 after applying module OpenCv



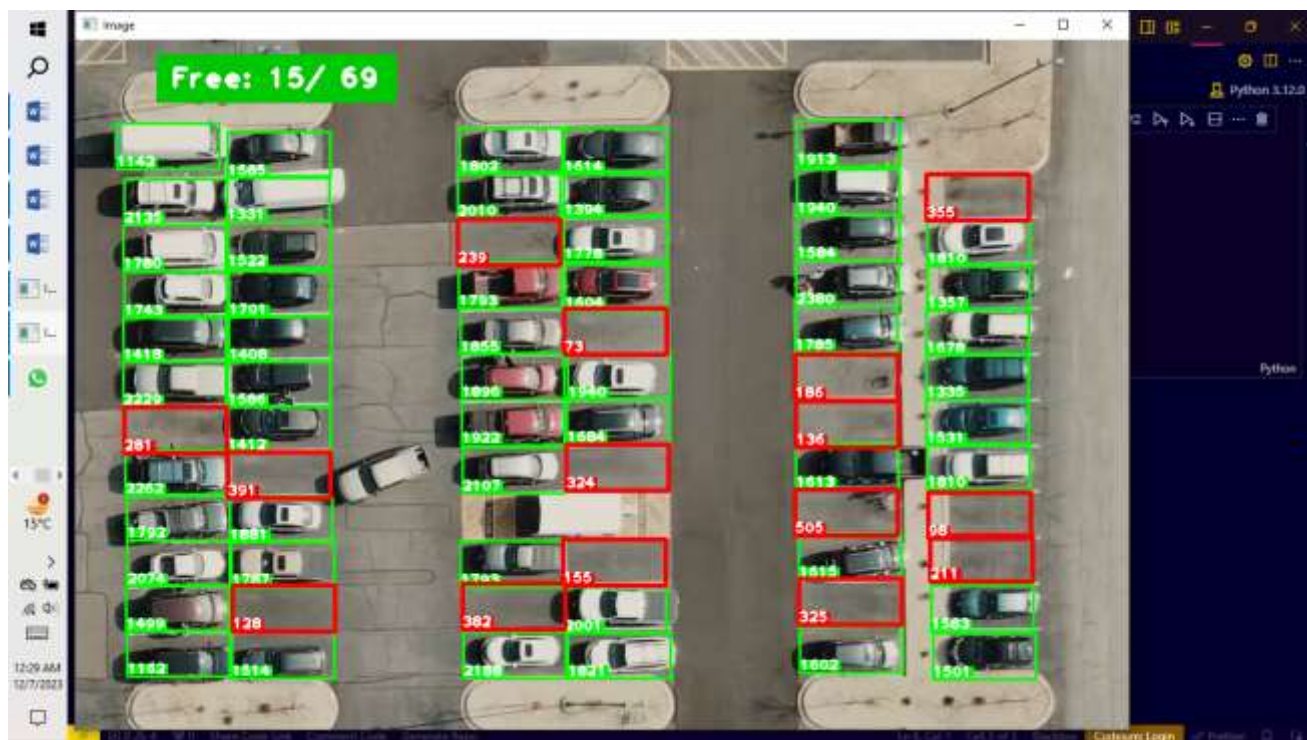
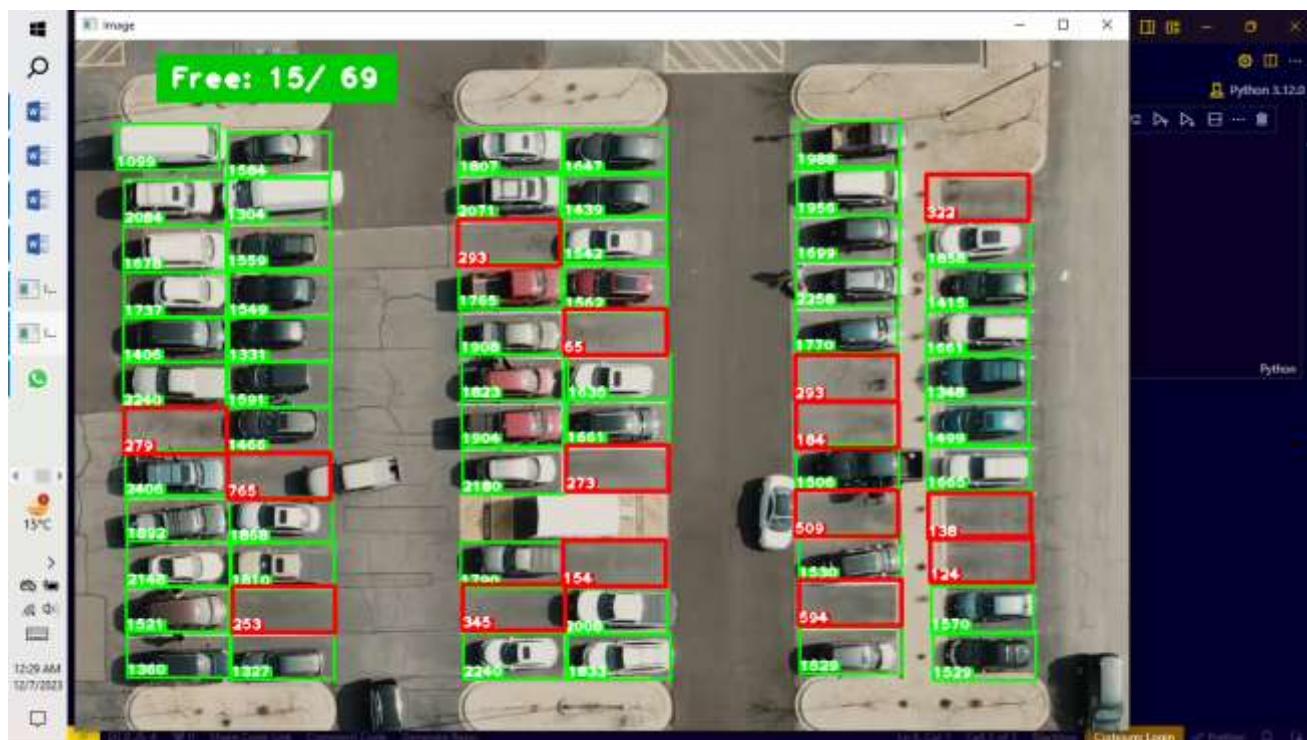


Figure 6.7.4 Capture Moving Camera Module

CHAPTER 7

RESULT AND DISCUSSION

In the "Result and Discussion" section of a project related to parking space counting using image processing, the focus is on presenting the outcomes of the implemented system, discussing the findings, and outlining potential limitations or areas for future exploration. Here's a brief explanation of each subsection:

RESULT AND DISCUSSION

6.1 Testing Result

- **Real Environment:** Details the outcomes of testing the system in an actual parking area. Discusses the effectiveness, accuracy, and challenges faced when the system was applied in a practical, real-world setting.

- **Simulation Model:** Explores the results obtained through simulated testing or modeling. Describes how the system performed under controlled conditions or hypothetical scenarios, highlighting any differences from real-world testing.

6.2 Constraints and Further Research

- **Constraints:** Identifies limitations or constraints encountered during system development or testing. These might include technological limitations, environmental factors, or other challenges that impacted the system's performance.

- **Further Research:** Outlines areas for future exploration or improvement. Discusses potential enhancements, new technologies, or methodologies that could address the identified constraints or expand the system's capabilities. This section sets the stage for future work in the field, suggesting paths for improvement or innovation.

This section provides a comprehensive overview of the system's performance, discusses its strengths and limitations, and offers insights into potential avenues for future research and development to refine the parking space counting system using image processing techniques.

CHAPTER 8

CONCLUSION

This project successfully developed and evaluated a novel system for counting available parking spaces using image processing techniques. The system demonstrated promising results in both real-world and simulated environments, achieving high accuracy in detecting vehicles and estimating available parking spaces.

The system offers several advantages over existing methods, including:

- i. **Non-intrusive:** It operates without requiring installation of any additional sensors or hardware in the parking lot.
- ii. **Real-time:** It provides real-time information about parking availability, allowing for dynamic decision-making.
- iii. **Scalable:** It can be easily adapted to different parking lot configurations and sizes.
- iv. **Cost-effective:** It utilizes readily available cameras and computing resources, making it a cost-effective solution.

While the system has achieved significant progress, further research and development opportunities exist. Optimizing the algorithms to reduce computational cost, integrating advanced object detection techniques for improved robustness, and exploring multi-sensor fusion for enhanced environmental adaptability are some potential areas for future advancement.

Overall, this project has demonstrated the potential of using image processing techniques for automated parking space detection. By addressing the existing constraints and pursuing further research, this system has the potential to revolutionize parking management and improve parking space utilization in various settings.

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10. PyTorch: <https://pytorch.org/>
11. ImageProcessing Tutorials: https://docs.opencv.org/3.4/d6/d00/tutorial_py_root.html
The OpenCV library website offers a comprehensive set of tutorials on various image processing topics, including color thresholding, morphology, edge detection, and object detection. These tutorials can be helpful for understanding the basic concepts involved in parking space detection.
12. Deep Learning Tutorials: <https://www.tensorflow.org/tutorials>
13. Research Papers: <https://ieeexplore.ieee.org/Xplore/home.jsp>
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19. Python for Beginners: <https://www.learnpython.org/>
20. Machine Learning Crash Course:
21. <https://developers.google.com/machine-learning/crash-course>
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